Olle Marmenlind

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Final-year Computer Engineering student at KTH with internships in embedded and full-stack engineering and experience as Technical Product Owner at Sevan AB. Primary interests include backend and systems engineering. Available for a software engineering internship or thesis project from early 2027, ahead of graduation in June 2027.

Education

Royal Institute of Technology (KTH) — Stockholm, Sweden
B.Sc. in Computer Engineering — **Expected June 2027**

- **Relevant Coursework:** Data Structures & Algorithms, Distributed Systems, Operating Systems, Networks, Databases; graduate-level Embedded Systems and Network Security
- **Certifications:** CCNA, CCNP

Engineering Experience

Motillo AB — Stockholm, Sweden
Software Engineer Intern — **March 2026 – June 2026**

- Reduced query times from 330 ms to ∼20 ms over 4 million rows by combining PostgreSQL materialized views for historical rollups with live data.
- Built a React/TypeScript control interface for UptimeRobot monitors and Litium order ingestion with Entra ID authentication and support for legacy Chrome 76 kiosks.
- Built an API for Elasticsearch order data with asynchronous ingestion using Hangfire and Polly to handle upstream rate limits, and OpenTelemetry for observability.

Proximion AB — Stockholm, Sweden
R&D Systems Engineer Intern — **October 2025 – January 2026**

- Built a multithreaded FreeRTOS pipeline with non-blocking IPC and an MCU-to-PC binary protocol using state-machine framing, sequence tracking, and CRC validation.
- Wrote kernel-level hooks to detect stack overflows and heap exhaustion, extracting thread-specific crash states through out-of-band hardware signaling without a debugger.
- Configured cross-platform GNU Make builds and self-hosted Linux CI/CD using hermetic Docker containers for reproducible builds and automated releases.

Other Professional Experience

Sevan AB — Stockholm, Sweden (Remote)
Technical Product Owner — **March 2022 – Present**

- Promoted to Product Owner, overseeing B2C platform growth from 4.5K to 200K users and migration of 300+ B2B accounts to e-commerce.
- Directed the roadmap, specified API schemas and data models, and coordinated development across external teams and vendors.

Selected Projects — continued on page 2

- **Token Catalog:** Built a Go pipeline and live web service providing versioned, verified Material Design token data for developers and coding agents.
- **Linduistics (WIP):** Developing an application that compares pronunciation and word meanings across five Germanic languages.

Technical Skills

Languages: Java, C++, Go, Python, TypeScript, C#
Systems & Tools: Linux, FreeRTOS, Docker, CI/CD, GNU Make, Git, TCP/UDP, OpenTelemetry
Web & Data: React, HTMX, Spring Boot, ASP.NET Core, REST, PostgreSQL, Elasticsearch, ETL

Selected Projects — continued

Linduistics Work in progress

Stack: Python, Go, PostgreSQL, React, TypeScript, spaCy, sentence-transformers

Developing a full-stack application for comparing vocabulary, pronunciation, and meaning across five Germanic languages through a searchable lexicon, text analysis, and language games.

- Built a pronunciation similarity engine using weighted edit distance, with sound substitution costs based on articulatory features. Used a precomputed distance matrix, Numba JIT compilation, and two-row buffering to reduce working memory from quadratic to linear.
- Built a cross-language dataset from Wiktionary translations and words with similar spellings, using multilingual embeddings to match specific word meanings. Retained source references and etymology while keeping ambiguous matches explicit.
- Refactored the dataset pipeline with field-level checks against an accepted baseline of 1.6 million word comparisons, preserving scores and classifications while allowing interrupted runs to reuse completed stages.
- Published comparisons and related lookup tables in one PostgreSQL transaction to prevent partially applied dataset updates.
- Built a React/TypeScript interface with a Go API and Python NLP service, including an interactive 2D/3D map that positions languages from pairwise distances using multidimensional scaling and iterative stress minimization.

Token Catalog tokencatalog.dev

github.com/sojupie/tokens

Stack: Go, SQLite, Chromium/chromedp, templ, HTMX, Docker

Built and deployed a service to make Material Design 3 token values directly usable in custom web components and agent-assisted development, after encountering friction accessing the data while styling a React application.

- Built the templ/HTMX interface with colors, typography, and shapes drawn from the extracted tokens across six light/dark and contrast combinations. Added filtered JSON exports and read-only MCP tools for developers and agents.
- Built a concurrent Go crawler using Chromium's DevTools Protocol to capture dynamically loaded token data. Tracked network activity and request completion to detect when capture was finished, with caching and retries for failed pages.
- Compared token values and their referenced tokens across platforms and themes to detect upstream design changes, ignoring changes to generated IDs and record ordering.
- Saved immutable captures and checked their coverage before publication, tying each review to the exact files through hashes. Rejected outdated releases and switched published datasets atomically, with recovery after interrupted updates.
- Reduced memory use when validating captured token data by decoding JSON and checking its schema one record at a time.
- Loaded the published SQLite index into memory at server startup so token requests require no database queries or calls to the upstream site.
- Achieved 100/100 in performance, accessibility, best practices, and SEO in a mobile PageSpeed Insights lab audit (11 October 2026). Lighthouse 13.5 emulated a Moto G Power with simulated Slow 4G (150 ms RTT, 1.64 Mb/s throughput) from Europe: 1.2 s largest contentful paint, 0 ms total blocking time, and zero cumulative layout shift.
- Implemented OpenAPI, Markdown responses, and MCP discovery, passing all 4 PageSpeed agentic browsing checks. Cloudflare's Is It Agent Ready? audit awarded 100/100 for discoverability, content, and bot access control. The overall score was 80/100 because the public, read-only service omits authentication and registration.